

MANUAL HYPERPARAMETER SELECTION FORM

2D CNN Model — Dogs vs. Cats Image Classification

Name: Jo Prisse T. Olfindo

Program: Doctor in Information Technology

Use this form to record, justify, and approve each hyperparameter value chosen by hand before a training run. The "Notebook Default" column shows the value currently used in Loading_Dataset_and_CNN.ipynb for reference — enter the value actually used for this run under "Chosen Value", even if it matches the default.

1. Project / Experiment Information

Project / Experiment Title	Manual Hyperparameter Selection and Final Evaluation of a 2D CNN for Dogs vs. Cats Image Classification
Model Architecture	2D Convolutional Neural Network (Conv2D)
Dataset Used	Kaggle Dogs vs. Cats images (Dog/ and Cat/ folders)
Notebook / File Reference	Loading_Dataset_and_CNN(6).ipynb (final); Trials 1-4 used for manual selection
Prepared By	Jo Prisse T. Olfindo
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Experiment / Trial No.	Trials 1-5; Final Model = Trial 5

2. Data Preprocessing Parameters

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification
Image Size (IMG_SIZE, square)	<i>e.g. 50 / 80 / 100 / 128 / 150</i>	100	128	Maximum allowed size. It produced better validation performance than the 100x100 baseline.
Color Mode	<i>Grayscale / RGB</i>	Grayscale	Grayscale	Retained from the notebook and used consistently across all trials.
Pixel Normalization	<i>X/255.0 / none</i>	<i>X/255.0</i>	X / 255.0	Scales pixel values from 0-255 to 0-1 for stable CNN training.
Shuffle Training Data	Yes / No	Yes (<i>random.shuffle</i>)	Yes; seed = 42	The data were shuffled. Trial 5 used a fixed seed for reproducibility.
Held-out Test Set (separate from training)	Yes / No	No (<i>validation_split only</i>)	Yes; 10% test set	Trial 5 reserved 2,500 images for testing and excluded them from model.fit().
Validation Split (of training set)	<i>0.0 - 0.3</i>	0.1	0.20 of remaining 90% (18% overall)	Final split: 72% training, 18% validation, and 10% testing.

3. Model Architecture Hyperparameters

Final Selected Configuration (based on Trials 1-4; evaluated in Trial 5)

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Filters	16 / 32 / 64 / 128	64	2 (retained from notebook)	Kept unchanged so the trials isolated image size and epoch effects.	82.64%	79.49%
Conv2D Layer 1 – Kernel Size	(3,3) / (5,5)	(3,3)	(3,3)	Retained from the notebook across all trials.		

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Conv2D Layer 1 – Activation	<i>relu / tanh / elu / leaky_relu</i>	<i>relu</i>	ReLU	Retained from the notebook across all trials.	82.64%	79.49%
Pooling Layer 1 – Type & Pool Size	<i>MaxPooling2D / AveragePooling2D; (2,2) / (3,3)</i>	<i>MaxPooling2D, (2,2)</i>	MaxPooling2D, (2,2)	Retained across all trials to reduce feature-map size.		
Dense (Fully Connected) Units	<i>16 / 32 / 64 / 128</i>	<i>64</i>	64, 64	Retained from the notebook; dense units were not tuned.		
Dropout Rate	<i>0.0 - 0.5</i>	<i>None (not used)</i>	0.0 (not used)	Dropout was not added so the architecture stayed consistent during tuning.		
Output Layer Activation	<i>sigmoid (binary) / softmax (multi-class)</i>	<i>sigmoid</i>	Sigmoid	Used for binary Dog/Cat classification.		

4. Training Hyperparameters

Hyperparameter	Options / Range Considered	Notebook Default	Chosen Value	Justification	Training Accuracy	Validation Accuracy
Loss Function	<i>binary_crossentropy / categorical_crossentropy</i>	<i>binary_crossentropy</i>	binary_crossentropy	Appropriate for two-class classification with a sigmoid output.	82.64%	79.49%
Optimizer	<i>adam / sgd / rmsprop</i>	<i>adam</i>	Adam	Used consistently across the trials; optimizer changes were not part of the comparison.		
Learning Rate	<i>0.1 / 0.01 / 0.001 / 0.0001</i>	<i>0.001 (Adam default)</i>	0.001 (Adam default)	Retained as the Adam default and not manually tuned.		
Batch Size	<i>8 / 16 / 32 / 64 / 128</i>	<i>32</i>	128	Used consistently across all trials and was manageable in the training environment.		
Number of Epochs	<i>e.g. 10 - 200</i>	<i>200</i>	7	Longer runs overfit. The best validation behavior occurred around 7 epochs.		
Early Stopping	<i>Yes (patience = ___) / No</i>	<i>No</i>	No	Overfitting was identified manually, then the final epoch count was fixed at 7.		

4.1. Summary of Manual Trials and Observations

Trial	Configuration	Training Acc.	Final Val. Acc.	Best Val. Acc.	Observation
1	100x100; 5 epochs	79.66%	77.20%	77.20%	Baseline. Small train-validation gap and no severe overfitting.
2	100x100; 50 epochs	98.53%	75.98%	78.58%	Severe overfitting. Final validation loss increased to 1.7252.
3	128x128; 20 epochs	95.56%	77.00%	80.26%	Higher resolution improved peak validation accuracy, but overfitting appeared after about 7-8 epochs. Final validation loss was 0.9997.

Trial	Configuration	Training Acc.	Final Val. Acc.	Best Val. Acc.	Observation
4	128x128; 7 epochs	82.72%	79.96%	79.96%	Best tuning balance. Validation loss was 0.4361 with a small train-validation gap.
5	128x128; 7 epochs; 72/18/10 split	82.64%	79.49%	79.49%	Final evaluation. Test accuracy 77.68%, macro F1 0.7763, ROC-AUC 0.8584.

Summary observation: More epochs did not improve generalization. Trials 2 and 3 overfit. Increasing the image size to 128x128 improved peak validation performance, and 7 epochs gave the most balanced result. Trial 5 kept these settings and added a fixed train/validation/test split for final evaluation.

5. Evaluation Results (after training with chosen values) - Final Model

Metric	Value Obtained	Target / Acceptable Range
Training Accuracy	82.64%	Not specified in the form
Validation Accuracy	79.49%	Not specified in the form
Test Accuracy	77.68%	Not specified in the form
Test Loss	0.4940	Not specified in the form
Precision / Recall / F1 (macro avg)	Precision 0.7795; Recall 0.7768; F1 0.7763	Not specified in the form

5.1. Paste here all screenshots/figures of the tuned model

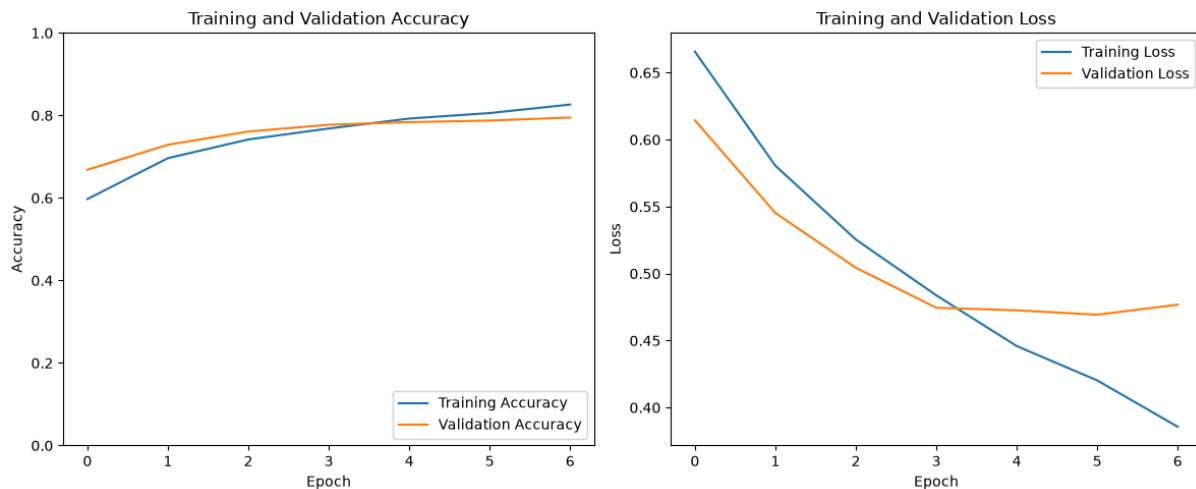


Figure 1. Training and validation accuracy/loss curves for the final 128x128, 7-epoch model.

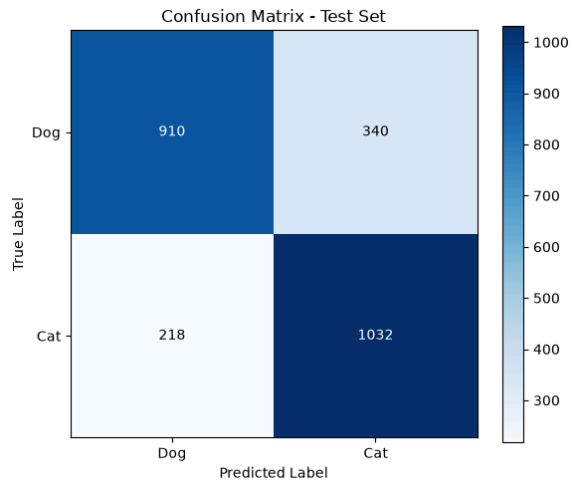


Figure 2. Confusion matrix on the 2,500-image test set.

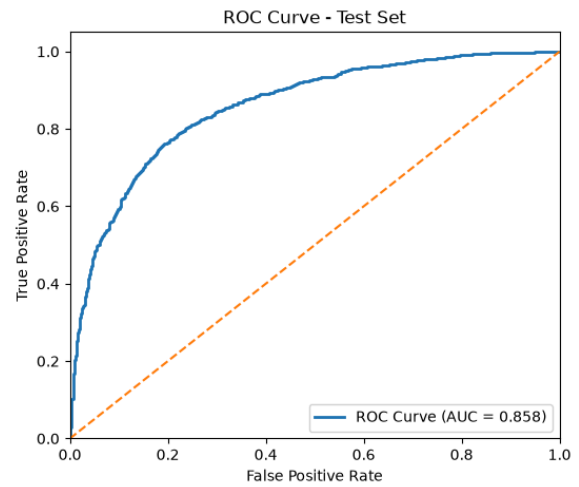


Figure 3. ROC curve on the final test set (AUC = 0.8584).

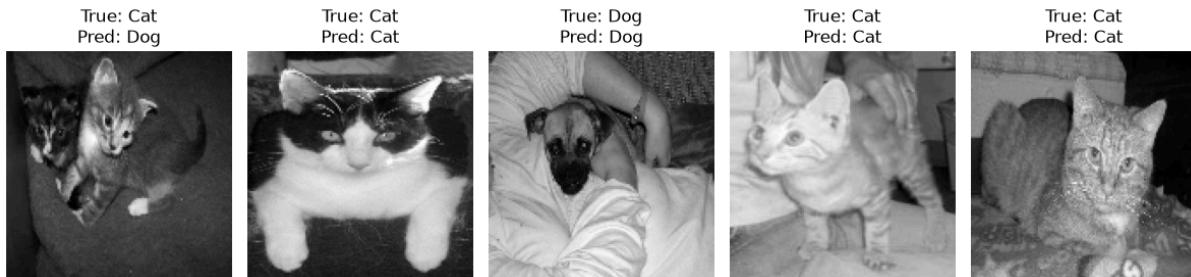


Figure 4. Example predictions using unseen samples from the final test set.

6. Overall Justification / Remarks

Summarize why this combination of hyperparameters was chosen (e.g., prior trial results, literature basis, computational constraints) and any observations (overfitting, underfitting, convergence issues).

The manual trials showed that simply increasing epochs did not improve generalization.

- Trial 1 (100x100, 5 epochs) was the baseline.
- Trial 2 (100x100, 50 epochs) reached 98.53% training accuracy but only 75.98% final validation accuracy, showing severe overfitting.
- Trial 3 used 128x128 images and reached the highest single validation accuracy of 80.26% around epoch 7, but continuing to 20 epochs again caused overfitting.
- Trial 4 therefore used 128x128 images for 7 epochs and achieved 82.72% training accuracy and 79.96% validation accuracy with a much smaller gap.
- Trial 5 kept the selected 128x128 image size and 7 epochs, used seed 42, and applied a 72% training / 18% validation / 10% test split.

The final model achieved 82.64% training accuracy, 79.49% validation accuracy, and 77.68% test accuracy. Test loss was 0.4940, macro precision 0.7795, macro recall 0.7768, macro F1-score 0.7763, and ROC-AUC 0.8584. The final choice was 128x128 and 7 epochs because it gave the best balance between validation performance and reduced overfitting.

Final test-set detail: 2,500 images; confusion matrix = [[910, 340], [218, 1032]]. The model correctly classified 1,942 of 2,500 test images.